

Operational Intelligence Evidence Pack

Benchmark rubric, review model, control comparisons, and falsification criteria for Operational Intelligence.

Evidence Question

What evidence would convince another experienced engineer that this operating model is useful? Operational Intelligence becomes stronger when claims can be tested, challenged, compared, and revised.

Review path

Structured practitioner review is captured through /contact so SRE, architecture, AI engineering, governance, executive, and product reviewers can challenge the doctrine with specific evidence requests.

Version Discipline

The Canonical Doctrine and Reference Architecture should evolve as versioned reference assets. Future v1.x changes should be incremental, evidence-based, and backward-compatible unless review or benchmarks prove that a term, layer boundary, or invariant is actively misleading.

Evidence Classes

Class	Purpose	Strong signal
Implementation example	Show end-to-end behavior	Public-safe inputs, outputs, states, and decisions
Benchmark fixture	Make behavior repeatable	Prompt, context, expected behavior, refusal, citation
Control comparison	Separate value from baselines	Dashboard-only, chatbot-only, ticket-only, and OI workflow compared
Practitioner review	Test real-world comprehension	Structured review by SRE, architect, AI engineer, governance, executive
Regression history	Show maturity over time	Failures tracked and corrected
Negative case	Test boundaries	Refuses confidential, unsupported, or unsafe requests

OI-ROOM-001 Benchmark Rubric

Dimension	Acceptance question
Retrieval	Did the system retrieve doctrine, reference architecture, and relevant OI-ROOM-001 context?
Grounding	Are all material claims supported by approved public content?
Evidence attribution	Are observation, inference, contradiction, missing evidence, and confirmed fact separated?
Hypothesis lifecycle	Do hypotheses move through explicit states?
Contradiction handling	Does contradictory evidence change confidence or route?
Decision safety	Are actions framed as reviewable options?
Operator control	Are consequential actions gated by explicit human approval?

Dimension	Acceptance question
Replay reproducibility	Can the scenario be replayed from approved context?
Refusal	Does the system refuse confidential or unsupported questions?

Evidence Ledger

Claim	Classification	Result	Next improvement
OI focuses on the decision path from evidence to action.	Original synthesis	Mixed	Collect SRE and observability practitioner feedback
The ten-layer model can guide implementation.	Original synthesis	Not independently tested	Review a second public-safe implementation example
Evidence graphs improve reviewability.	Derived	Plausible	Compare against chatbot-only and dashboard-only explanations
Replay seeds support regression testing.	Derived	Partial	Add replay-backed workflow fixtures
Operator control should gate consequential actions.	Established governance principle applied to OI	Strong	Add approval class examples

Minimum Conformance Checklist

Requirement	Observable proof	Failure signal
Evidence before conclusion	Recommendations cite approved evidence or missing evidence.	Fluent RCA with no source trail.
Transaction context	Affected customer, process, or business journey is identified.	Local metric treated as the whole incident.
Topology boundary	Dependency, ownership, blast radius, and freshness are visible.	Stale service map treated as fact.
Evidence typing	Observation, inference, contradiction, missing evidence, and fact are separated.	Contradictions hidden in prose.
Hypothesis lifecycle	Competing hypotheses move through explicit states.	First plausible explanation becomes final answer.
Evaluation gate	Grounding, citations, refusal, contradictions, and action safety are tested separately.	Single aggregate score implies trust.
Decision packet	Action includes risk, reversibility, owner, fallback, and approval class.	Assistant recommends action without review context.
Operator control	Approval, escalation, override, and refusal boundaries are explicit.	Automation boundary is hidden.
Replay seed	Scenario can be reproduced from approved context and expected behavior.	Demo cannot be rerun after changes.
Learning loop	Reviewed outcome updates memory, patterns, docs, and fixtures.	Learning remains unstructured narrative.

Control Comparison Template

Criterion	Dashboard-only	Chatbot-only	Operational Intelligence
Evidence trail	Signals visible but reasoning is external	Often summarized without stable receipts	Claims cite typed evidence and unknowns
Hypotheses	Human-maintained outside the tool	May collapse to first plausible answer	Lifecycle states and falsification criteria are visible
Contradictions	Require manual noticing	May be hidden by fluent narrative	Contradictory evidence is first-class
Action safety	Handled by process outside dashboard	Risk of over-suggesting action	Decision packet names risk, owner, and approval class
Learning	Postmortem and memory are separate	Conversation history is weak memory	Reviewed outcome becomes replay, fixture, and memory

OI-ROOM-001 Control Comparison Protocol

Use the same synthetic facts, red herrings, contradictory evidence, and missing-evidence condition across dashboard-only, chatbot-only, ticket-only, and Operational Intelligence workflow runs. This is a measurement design, not a claim that superiority has already been proven.

Mode	Allowed input	Expected output	Primary risk
Dashboard-only	Synthetic telemetry, traces, status signals	Human investigation notes	Visibility without reasoning trace
Chatbot-only	Unstructured public-safe case context	Narrative explanation with citations	Fluency without durable evidence state
Ticket-only	Synthetic ticket, comments, resolution notes	Timeline and summary	Record without live hypothesis movement
OI workflow	Typed evidence, transaction path, topology, hypotheses, eval gates, approval boundary	Decision packet, uncertainty, replay seed, approval class	Slower if structure does not improve judgment

Score each mode on evidence completeness, hypothesis quality, contradiction handling, missing-evidence honesty, decision safety, replayability, and reviewer confidence. Report results as evidence ledger entries with limitations.

Reviewer-Run Worksheet

Step	Reviewer action	Required evidence
1	Use same OI-ROOM-001 facts for every mode	Facts, red herring, contradiction, missing evidence, timing, approval boundary
2	Inspect dashboard-only output	Visible signals and operator inferences
3	Inspect chatbot-only output	Citations, uncertainty, unsupported claims, action language
4	Inspect ticket-only output	Timeline quality, ownership, replayability
5	Inspect OI workflow output	Evidence graph, hypothesis state, eval gates, decision packet, approval class
6	Record evidence ledger entry per dimension	Verdict, limitation, failure signal, next improvement

Verdict	Use when
Strong	Another reviewer can reproduce the judgment from exposed evidence.
Mixed	Useful information remains but a key state, boundary, source, or uncertainty is lost.
Weak	The output relies on private memory, unsupported inference, hidden assumptions, or unreviewable action.
Not assessable	The mode does not expose enough information to judge.

Practitioner Review Rubric

Reviewer	Question they should be able to answer	Weak signal
SRE	Where does the model improve investigation without replacing incident command?	Cannot distinguish OI from observability or AIOps
Architect	Can two teams implement the contracts similarly?	Layer boundaries are interpreted incompatibly
AI engineer	Which gates prevent unsafe model behavior?	Only aggregate scores or confidence prose are shown
Governance reviewer	Where are approval, audit, refusal, and data boundaries enforced?	Autonomy scope is implicit
Executive	What evidence would justify further investment?	Only narrative claims, no benchmark or feedback path

Weakening Conditions

The doctrine should be revised if experienced SRE teams cannot distinguish it from existing practice, if independent teams interpret the layers incompatibly, if evidence graphs add complexity without clarity, or if evaluation gates fail to catch obvious regressions.